Deferred-Quotient Vacuity in BN254 Wrappers: A Repair Note on Algebraic Exactness

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Abstract

We study BN254 wrapper arithmetizations for computations over the 31-bit Mersenne prime field $\mathbb{F}_p$ with $p = 2^{31} - 1$. In this setting, modular reduction is encoded by deferred-quotient row constraints of the form $L = D + pQ$ over the outer field. This note proves only a restricted-model rowwise exactness theorem for such wrappers; it does not claim full backend or Fiat–Shamir soundness. The core pathology is outer-field generic: when p is invertible in the outer field, an unbounded quotient witness makes the rowwise equality $L = D + pQ$ vacuous.

We formalize this vacuity at row and polynomial levels. Quotient range binding excludes the vacuous p^{-1}-witness class. Canonical residue binding together with no-wrap hypotheses then upgrades a satisfied field equality $L = D + pQ$ to an exact integer identity in $\mathbb{Z}$; on designated residue rows this recovers unique Euclidean quotient/remainder semantics.

We instantiate the theorem for a concrete BN254 family set with one-limb 31/66-bit quotient classes and explicit familywise no-wrap inequalities. This paper is a theorem-level repair note for the algebraic exactness layer of this wrapper class.

1 Introduction

What goes wrong. In the wrapper class considered here, modular reduction is encoded by enforcing $L = D + pQ$ over the outer field $R = \mathbb{F}_r$. If Q is left unbounded, this equality is algebraically vacuous: the prover can always choose $Q = p^{-1}(L - D)$ in R, which satisfies the field equation without establishing intended integer-division semantics.

Why it matters. In this wrapper class, soundness arguments read $L = D + pQ$ as an integer statement with a canonical remainder and quotient. Without explicit canonical range binding and no-wrap control, that interpretation is invalid.

What we prove. We formalize vacuity at row and polynomial levels and then separate two repair steps. First, quotient range binding excludes the vacuous $Q = p^{-1}(L - D)$-witness pattern. Second, canonical residue binding together with audited no-wrap bounds upgrades bounded field equalities $L = D + pQ$ to exact identities in $\mathbb{Z}$, so designated residue rows recover Euclidean quotient/remainder semantics. The contribution is theorem-level in cryptographic arithmetization: we isolate the soundness-reading gap and prove an instantiated exactness repair under explicit canonical-binding premises.

Repair target. This note isolates and repairs the algebraic exactness gap in a concrete BN254/M31 deferred-quotient wrapper design. The repaired object is the normalized non-negative, one-limb deferred-quotient family instantiated in the appendices: the compression identity is unchanged, but exact $\mathbb{Z}$-semantics now rests on explicit quotient/residue binding and no-wrap premises. The present note is self-contained for the theorem proved here.

What we do not prove. This note does not prove backend transcript extraction, PCS-level cryptographic soundness, or Fiat–Shamir closure.

1.1 Formal Setup

Let

$$p := p_{\mathrm{M31}} = 2^{31} - 1, \qquad R := \mathbb{F}_r,$$

where R is the BN254 scalar field. We assume the one-limb headroom condition

$$p \cdot 2^{66} + p < r.$$

For $p = 2^{31} - 1$, the left-hand side equals

$$(2^{31} - 1)(2^{66} + 1) = 2^{97} + 2^{31} - 2^{66} - 1 < 2^{97},$$

and the BN254 scalar modulus satisfies $r > 2^{253}$. Hence $p \cdot 2^{66} + p < r$. This class-level headroom condition is sufficient for the one-limb 31/66-bit quotient classes used below. Full family-specific left/right no-wrap bounds are discharged in Appendix B.

Let

$$\Omega = \{\omega^0, \omega^1, \ldots, \omega^{m-1}\} \subset R$$

be the public row domain indexing the target transcript tables, with size

$$m := |\Omega|.$$

Let $\mathcal{I}$ be the finite index set of scalar local relations aggregated by the compressed wrapper check.

We keep the deferred-quotient compression layer unchanged. For each scalar local constraint index $t \in \mathcal{I}$, let

$$L_t(X), D_t(X) \in R[X]$$

denote the (potentially high-degree) *composition polynomials* induced by the instantiated row templates, i.e., obtained by evaluating the row-template polynomials on the underlying degree-$< m$ witness polynomials over Ω. Let

$$Q_t(X) \in R[X]$$

denote the low-degree quotient witness polynomial (so $\deg Q_t < m$) that interpolates the quotient row table on Ω. Define the aggregated numerator

$$N(X) := \sum_{t \in \mathcal{I}} \lambda_t \big(L_t(X) - D_t(X) - pQ_t(X)\big),$$

where λ_t are random aggregation coefficients sampled by the wrapper verifier (or via Fiat–Shamir in a non-interactive instantiation). Let

$$Z_\Omega(X) := \prod_{u \in \Omega} (X - u)$$

and let $H(X) \in R[X]$ denote the prover-supplied quotient witness for the claim that $N(X)$ is divisible by $Z_\Omega(X)$. Whenever N vanishes on Ω, define the *true* quotient

$$H_{\mathrm{true}}(X) := \frac{N(X)}{Z_\Omega(X)}.$$

Thus H_{true} is the exact quotient implied by the algebraic relation, while H is the witness polynomial supplied in the compressed check.

Composition versus interpolant. We emphasize that $L_t(X)$ and $D_t(X)$ above are *not* defined as the unique degree-$< m$ interpolants of the rowwise values $(L_t[j])_{j\in\Omega}$ and $(D_t[j])_{j\in\Omega}$. Under quadratic templates, the composition polynomials can have degree up to $2m-2$, so N is genuinely high-degree and H_{true} need not be identically zero on honest transcripts.

Contributions. This paper makes three main contributions:

(i) an outer-field generic diagnosis of deferred-quotient vacuity;

(ii) a canonical-lift exactness theorem that upgrades bounded field equalities $L = D + pQ$ to exact integer identities;

(iii) a BN254 instantiation with explicit 31/66-bit quotient classes and familywise no-wrap inequalities.

Supplementary implementation evidence appears in Appendix D.

1.2 Instantiated model

The theorem is proved for the explicit model used in this manuscript: normalized non-negative row templates, unsigned quotient classes, designated canonical residue rows, and the instantiated family set $\{\mathrm{inv}, \mathrm{EF}, \mathrm{EE}, \mathrm{hor}, \mathrm{car}\}$. For the hor and car families, the theorem is stated for the finite template catalogue instantiated in Appendices B and C.

Proposition 1.1 (The deferred-quotient vacuity pathology is outer-field generic)**.** *Let F be any field, let $p \in F^\times$, and let*

$$\Omega = \{\omega^0, \omega^1, \dots, \omega^{m-1}\} \subset F$$

be a finite row domain. For arbitrary rowwise values

$$(L[j], D[j])_{j=0}^{m-1} \subset F^2,$$

define

$$Q[j] := p^{-1}\big(L[j] - D[j]\big) \in F \quad \text{for } j = 0, \dots, m-1,$$

and let $Q(X)$ be the unique degree-$< m$ interpolant through $(Q[j])_{j=0}^{m-1}$. Then

$$L[j] - D[j] - pQ(\omega^j) = 0 \quad \text{for all } j = 0, \dots, m-1.$$

Hence, absent additional range binding or canonical-lift premises, the rowwise identity $L = D + pQ$ carries no intended integer-division semantics over any outer field in which p is invertible.

Proof. For each j,

$$pQ(\omega^j) = pQ[j] = L[j] - D[j]$$

by the definition of the row values $Q[j]$ and the interpolant $Q(X)$. Therefore

$$L[j] - D[j] - pQ(\omega^j) = 0$$

for every row j. □

Remark 1.2 (Role of the BN254 specialization)**.** *Proposition 1.1 shows that the vacuity phenomenon treated here is not specific to BN254. The BN254 specialization enters only through the concrete one-limb headroom inequalities, the chosen 31/66-bit quotient classes, and the instantiated family audit constants used by the repair.*

Instantiated family class. The claims proved here apply to the normalized non-negative, unsigned, designated-residue deferred-quotient family class explicitly instantiated in this manuscript, namely

$$\{\mathrm{inv}, \mathrm{EF}, \mathrm{EE}, \mathrm{hor}, \mathrm{car}\}.$$

Families whose satisfying assignments require signed quotient ranges, negative normalizations, or a different residue convention are not treated here.

1.3 Deferred-quotient vacuity in R

By Proposition 1.1, the deferred-quotient vacuity mechanism is outer-field generic whenever p is invertible; this subsection specializes the notation to $R = \mathbb{F}_r$ only to match the audited BN254 wrapper instance analyzed in this manuscript.

The deferred-quotient layer aims to represent rowwise Euclidean division by p by enforcing field equalities of the form

$$L_t(\omega^j) - D_t(\omega^j) = pQ_t(\omega^j)$$

for all $\omega^j \in \Omega$, where $Q_t(X)$ is the degree-$< m$ quotient witness polynomial interpolating the row-table values $(Q_t[j])_{j\in\Omega}$. Since p is invertible in R, this rowwise constraint on Ω is vacuous unless quotient values are additionally range-bound: at the level of actual composition evaluations $(L_t(\omega^j), D_t(\omega^j))_{j\in\Omega}$, the prover can set

$$Q_t[j] := p^{-1}\big(L_t(\omega^j) - D_t(\omega^j)\big) \in R$$

so that $L_t(\omega^j) - D_t(\omega^j) = pQ_t(\omega^j)$ holds on Ω. There is then a unique degree-$< m$ interpolant through the chosen quotient row values, so the rowwise constraint carries no intended integer-division semantics unless quotient cells are bound to small integer ranges.

Definition 1.3 (Compression/rowwise interface consistency)**.** *For each relation index t, let $(L_t[j], D_t[j], Q_t[j])_{j\in\Omega}$ be the rowwise values defined by the realization rule of Definition C.5 together with the zero-extension rule of Definition C.13. The wrapper compression layer uses the composition polynomials $L_t(X), D_t(X)$ from the instantiated templates (as in the numerator definition above), and the quotient witness polynomial $Q_t(X)$ interpolating $(Q_t[j])_{j\in\Omega}$. Consistency requires that*

$$L_t(\omega^j) = L_t[j], \qquad D_t(\omega^j) = D_t[j], \qquad Q_t(\omega^j) = Q_t[j] \quad \textit{for all } j \in \{0, \dots, |\Omega| - 1\}.$$

Under Definition 1.3, this rowwise vacuity extends directly to the polynomial quotient-check layer.

Proposition 1.4 (Polynomial-level vacuity under unbounded quotient witnesses)**.** *Fix a relation index t and a witness assignment determining composition polynomials*

$$L_t(X), D_t(X) \in R[X]$$

with

$$\deg L_t, \deg D_t \le 2m - 2.$$

Assume quotient witnesses are not range-bound. For each row $\omega^j \in \Omega$, define

$$Q_t[j] := p^{-1}\big(L_t(\omega^j) - D_t(\omega^j)\big) \in R,$$

let $Q_t(X)$ be the unique degree-$< m$ interpolant through the row values $(Q_t[j])_{j\in\Omega}$, and set

$$R_t(X) := L_t(X) - D_t(X) - pQ_t(X).$$

Then $R_t(\omega^j) = 0$ for all $j \in \{0, \dots, m-1\}$, so $Z_\Omega \mid R_t$. Moreover

$$\deg R_t \le 2m - 2,$$

hence the quotient

$$H_t(X) := \frac{R_t(X)}{Z_\Omega(X)}$$

has degree $\le m - 2$. Therefore the usual low-degree quotient-check layer alone does not enforce intended Euclidean-division semantics.

Proof. For each j,

$$pQ_t(\omega^j) = pQ_t[j] = L_t(\omega^j) - D_t(\omega^j)$$

by construction of the interpolated quotient values. Hence

$$R_t(\omega^j) = L_t(\omega^j) - D_t(\omega^j) - pQ_t(\omega^j) = 0$$

for all $\omega^j \in \Omega$, so $Z_\Omega \mid R_t$. The degree bound follows from

$$\deg L_t, \deg D_t \le 2m - 2, \qquad \deg(pQ_t) < m.$$

Since $\deg Z_\Omega = m$, the quotient degree bound $\deg H_t \le m - 2$ follows. □

Evaluation-level formulation. Proposition 1.4 is intentionally stated directly in terms of the actual composition evaluations $L_t(\omega^j), D_t(\omega^j)$. It therefore isolates the vacuity phenomenon without depending on any later row-table serialization convention.

Corollary 1.5 (Aggregated numerator vacuity under unbounded quotient witnesses)**.** *Let $\mathcal{I}$ be a finite relation-index set. For each $t \in \mathcal{I}$, assume the setup of Proposition 1.4, and define*

$$R_t(X) := L_t(X) - D_t(X) - pQ_t(X).$$

For coefficients $(\lambda_t)_{t \in \mathcal{I}} \subset R$, set

$$N(X) := \sum_{t \in \mathcal{I}} \lambda_t R_t(X).$$

Then $Z_\Omega \mid N$. If additionally

$$\deg L_t, \deg D_t \le 2m - 2, \qquad \deg Q_t < m \quad (\forall t \in \mathcal{I}),$$

then $\deg N \le 2m - 2$, and the quotient

$$H_N(X) := \frac{N(X)}{Z_\Omega(X)}$$

has degree $\le m - 2$.

Proof. For each t, Proposition 1.4 gives $Z_\Omega \mid R_t$. Hence Z_Ω divides every R-linear combination $\sum_t \lambda_t R_t$, proving $Z_\Omega \mid N$. The degree bound follows from $\deg R_t \le 2m - 2$ for each t, so $\deg N \le 2m - 2$, and then

$$\deg H_N \le (2m - 2) - m = m - 2.$$

□

Terminology. We refer to this pattern as a *deferred-quotient* wrapper: modular reductions are enforced via identities of the form $L = D + pQ$ in the outer field, deferring explicit non-native division to a range-binding layer.

Toy counterexample (p^{-1} witness). At the row level, if the circuit only enforces $L = D + pQ$ in R with an unbounded quotient witness, a malicious prover can satisfy the check while violating the intended modulo-p semantics. For instance, take $L = 0$ and $D = 1$ in R, and set $Q := -p^{-1} \bmod r$. Then $0 = 1 + p(-p^{-1})$ holds in R even though it does not encode any valid integer division statement.

1.4 Repair summary and claim level

The missing ingredient is a lightweight realization in this model of the canonical-lift property $\mathcal{P}_{\text{canon}}$: residue cells must be bound to their canonical integer ranges, and quotient/carry cells must be bounded as small integers, so that a field equality in R upgrades to an exact Euclidean-division statement in $\mathbb{Z}$.

Definition 1.6 (Canonical-lift property guarantee)**.** *We say a binding layer realizes the canonical-lift property guarantee $\mathcal{P}_{\text{canon}}$ if, for every satisfying assignment:*

(i) every designated residue cell is the image in R of an integer in $I_p = \{0, \ldots, p-1\}$;

(ii) every designated quotient/carry cell is the image in R of an integer in its declared small range (e.g., I_{31}, I_{66}, $\{0, 1\}$, or $\{0, \ldots, 7\}$);

(iii) for every satisfied deferred-quotient row relation $L_t[j] = D_t[j] + pQ_t[j]$ in R, the row values admit canonical integer lifts $\widehat{L_t[j]}, \widehat{D_t[j]}, \widehat{Q_t[j]}$ such that

$$0 \le \widehat{L_t[j]} < r, \qquad 0 \le \widehat{D_t[j]} + p\widehat{Q_t[j]} < r;$$

hence the relation holds as an exact identity in $\mathbb{Z}$, and on designated residue rows it encodes Euclidean division by p.

Concretely, this manuscript realizes $\mathcal{P}_{\text{canon}}$ by adding only:

(i) exact residue canonicality in $[0, p-1]$ using 31-bit limb decomposition plus one exclusion gate;

(ii) coarse 31-bit or 66-bit bounds for quotient witnesses, and tiny bounds for carries;

(iii) audited no-wrap bounds for all satisfying witnesses.

These yield the main exactness theorem, Theorem 1.14. We include only a brief remark on composing the theorem with a full wrapper system (Remark 1.15).

1.5 Positioning and related work

This construction follows a standard non-native (foreign-field) arithmetic pattern: express modular reduction via an equality carrying an explicit quotient/remainder witness, and then range-bind those witnesses to small integer intervals so the equality has a unique $\mathbb{Z}$-semantics. This pattern is ubiquitous in PLONKish arithmetizations and their lookup-based range-check extensions [1, 2, 3]. Recent work on foreign-field arithmetic in PLONK studies tight headroom/no-wrap premises and their constraint costs under such bounds [4]. For a representative engineering codebase that includes non-native arithmetic gadgets in Halo2 circuits, see [5].

Accordingly, we present a focused theorem-level treatment for deferred-quotient BN254 wrappers: our novelty is not a new generic foreign-field gadget, but a formal diagnosis of a specific algebraic vacuity class (the p^{-1}-witness failure mode) and a proof that explicit canonical-lift binding with audited no-wrap bounds restores exact integer semantics in the stated model.

1.6 Column classes

We partition the witness columns into five classes:

- $\mathcal{C}_p$: residue columns that must lie in

$$I_p := \{0, 1, \ldots, p-1\}.$$

- $\mathcal{C}_{31}$: quotient columns that only need the coarse bound

$$I_{31} := \{0, 1, \ldots, 2^{31} - 1\}.$$

- $\mathcal{C}_{66}$: quotient columns that only need the coarse bound

$$I_{66} := \{0, 1, \ldots, 2^{66} - 1\}.$$

- $\mathcal{C}_{\text{bit}}$: carry columns in $\{0, 1\}$.
- $\mathcal{C}_8$: small carry columns in $\{0, 1, \ldots, 7\}$.

The only exact canonicality requirement is for $\mathcal{C}_p$. For quotient columns, it is enough to enforce a public upper bound strictly below r/p so that, together with RHS no-wrap and canonical remainder binding on designated residue rows, any equality of the form

$$L = D + pQ \quad \text{in } R$$

upgrades to an exact identity in $\mathbb{Z}$. On those designated residue rows, this exact identity yields the unique Euclidean quotient and canonical remainder.

Definition 1.7 (Complete quotient coverage)**.** *Let $\mathcal{Q}$ be the set of all quotient witness columns used by the target deferred-quotient arithmetization. There is a public class map*

$$\kappa : \mathcal{Q} \to \{31, 66\},$$

with induced partition

$$\mathcal{Q}_{31} := \kappa^{-1}(31), \qquad \mathcal{Q}_{66} := \kappa^{-1}(66),$$

such that

$$\mathcal{Q}_{31} \cap \mathcal{Q}_{66} = \varnothing, \qquad \mathcal{Q}_{31} \cup \mathcal{Q}_{66} = \mathcal{Q}.$$

Every row-local quotient symbol $Q_t[j]$ is wired to exactly one column in $\mathcal{Q}$. Columns in $\mathcal{Q}_{31}$ are constrained by (4)–(5)*; columns in $\mathcal{Q}_{66}$ are constrained by* (6)–(7)*.*

Coverage/wiring metadata location. For readability, explicit quotient inventory and selector-level wiring metadata are moved to Appendix C.

1.7 Binding mechanism

We now define the minimal binding constraints that realize $\mathcal{P}_{\text{canon}}$.

1.7.1 Exact canonical residues in $[0, p-1]$

For every cell $x = W[j]$ with $W \in \mathcal{C}_p$, introduce two limb witnesses

$$x^{(0)}, x^{(1)} \in R$$

and one exclusion witness

$$\nu_x \in R.$$

The circuit enforces

$$x = x^{(0)} + 2^{16} x^{(1)}, \tag{1}$$

$$x^{(0)} \in T_{16} := \{0, 1, \ldots, 2^{16} - 1\}, \qquad x^{(1)} \in T_{15} := \{0, 1, \ldots, 2^{15} - 1\}, \tag{2}$$

and

$$(x - p)\nu_x = 1. \tag{3}$$

Equations (1)–(2) imply

$$0 \le x \le 2^{31} - 1 = p.$$

Equation (3) excludes the unique bad endpoint $x = p$, so $x \in I_p$.

Lemma 1.8 (Mersenne canonicality)**.** *If $x \in R$ satisfies* (1)–(3)*, then x is the canonical integer lift of a unique element of $\mathbb{F}_p$.*

Proof. From (1)–(2), we obtain $x \in [0, 2^{31} - 1]$. Since $2^{31} - 1 = p$, the only value in this interval that is not in $I_p = \{0, \ldots, p-1\}$ is $x = p$. Equation (3) implies $x \neq p$. Hence $x \in [0, p-1]$, so it is the canonical lift of a unique residue modulo p. □

1.7.2 Small quotient columns

For every cell $q = W[j]$ with $W \in \mathcal{C}_{31}$, introduce two limb witnesses

$$q^{(0)}, q^{(1)} \in R$$

and enforce

$$q = q^{(0)} + 2^{16} q^{(1)}, \tag{4}$$

$$q^{(0)} \in T_{16}, \qquad q^{(1)} \in T_{15}. \tag{5}$$

Hence

$$0 \le q < 2^{31}.$$

1.7.3 Large quotient columns

For every cell $q = W[j]$ with $W \in \mathcal{C}_{66}$, introduce five limb witnesses

$$q^{(0)}, q^{(1)}, q^{(2)}, q^{(3)}, q^{(4)} \in R$$

and enforce

$$q = q^{(0)} + 2^{16} q^{(1)} + 2^{32} q^{(2)} + 2^{48} q^{(3)} + 2^{64} q^{(4)}, \tag{6}$$

$$q^{(0)}, q^{(1)}, q^{(2)}, q^{(3)} \in T_{16}, \qquad q^{(4)} \in T_2 := \{0, 1, 2, 3\}. \tag{7}$$

Hence

$$0 \le q < 2^{66}.$$

1.7.4 Carry columns

For every cell $c = W[j]$ with $W \in \mathcal{C}_{\text{bit}}$, enforce

$$c(c-1) = 0. \tag{8}$$

For every cell $u = W[j]$ with $W \in \mathcal{C}_8$, enforce

$$u \in T_8 := \{0, 1, \ldots, 7\}. \tag{9}$$

1.7.5 Batched lookup realization

All instances of

$$x^{(0)} \in T_{16},\ x^{(1)} \in T_{15},\ q_{31}^{(0)} \in T_{16},\ q_{31}^{(1)} \in T_{15},\ q_{66}^{(0)} \in T_{16},\ q_{66}^{(1)} \in T_{16},\ q_{66}^{(2)} \in T_{16},\ q_{66}^{(3)} \in T_{16},\ q^{(4)} \in T_2,\ u \in T_8$$

are enforced using one batched native lookup layer in the Tier-2 PLONKish circuit. Thus the binding mechanism uses only:

- native BN254 lookups into tiny public tables T_{16}, T_{15}, T_2, T_8,
- one non-equality constraint $(x-p)\nu_x = 1$ per residue cell, and
- native linear recomposition constraints.

No foreign-field multiplication gadget, no CRT consistency check, and no multi-limb carry chain is introduced.

1.8 Why the p^{-1}-witness pathology is eliminated

The previous vacuity failure mode exploited the fact that a malicious prover could choose an unconstrained field element

$$Q = -p^{-1} \pmod{r}.$$

This is impossible once the quotient cells are bounded as small integers.

Lemma 1.9 (Bounded deferred-quotient exactness)**.** *Let $B \in \{2^{31}, 2^{66}\}$. Let $L, D, q \in R$ satisfy:*

(i) L is the image in R of an integer $\widehat{L}$ with

$$0 \le \widehat{L} < r;$$

(ii) D is the image in R of an integer $\widehat{D}$ with

$$0 \le \widehat{D} < r;$$

(iii) q is the image in R of an integer $\widehat{q} \in [0, B)$;

(iv) in $\mathbb{Z}$,

$$0 \le \widehat{D} + p\widehat{q} < r;$$

(v) in R,

$$L = D + pq.$$

Then

$$\widehat{L} = \widehat{D} + p\widehat{q} \quad \text{in } \mathbb{Z}.$$

If additionally $\widehat{D} \in I_p$*, then necessarily*

$$\widehat{D} = \widehat{L} \bmod p, \qquad \widehat{q} = \left\lfloor \frac{\widehat{L}}{p} \right\rfloor.$$

Proof. By (i) and (iv), both $\widehat{L}$ and $\widehat{D} + p\widehat{q}$ lie in $[0, r)$. Using (v),

$$L - (D + pq) = 0 \text{ in } R$$

upgrades to integer equality:

$$\widehat{L} = \widehat{D} + p\widehat{q}.$$

If $\widehat{D} \in I_p$, uniqueness of Euclidean division by p yields the final statement. □

Corollary 1.10 (Exclusion of the p^{-1}-witness class under quotient range binding)**.** *If a quotient cell is constrained by* (4)–(5) *or by* (6)–(7)*, then it cannot equal* $-p^{-1} \pmod{r}$*. Hence the specific* p^{-1}*-based vacuity pattern is excluded.*

Proof. Any such quotient cell is an integer $\widehat{q}$ in $[0, 2^{31})$ or in $[0, 2^{66})$. Assume for contradiction that

$$\widehat{q} \equiv -p^{-1} \pmod{r}.$$

Then

$$p\widehat{q} + 1 \equiv 0 \pmod{r}.$$

If $\widehat{q} < 2^{31}$, then

$$0 < p\widehat{q} + 1 \leq p(2^{31} - 1) + 1 < p \cdot 2^{31} + p < r.$$

If $\widehat{q} < 2^{66}$, then

$$0 < p\widehat{q} + 1 \leq p(2^{66} - 1) + 1 < p \cdot 2^{66} + p < r.$$

In either case $0 < p\widehat{q} + 1 < r$, so it cannot be divisible by r, a contradiction. Therefore $\widehat{q} \not\equiv -p^{-1} \pmod{r}$. □

1.9 Exact row soundness

For each active row relation, once a local realization is matched to an audited normalized template and all participating cells are canonically range-bound, we write

$$\widehat{L_t[j]}, \qquad \widehat{D_t[j]}, \qquad \widehat{Q_t[j]}$$

for the corresponding audited/canonical integer evaluations. Concretely, on active rows these are the integers supplied by Definition C.6 together with Lemma C.10; on inactive rows they are zero by Definition C.13. Their images in R are denoted

$$L_t[j], \qquad D_t[j], \qquad Q_t[j].$$

We enforce local relations as

$$L_t[j] = D_t[j] + pQ_t[j] \quad \text{in } R.$$

Notation convention for the inverse family. To avoid collision between the inverse-family multiplicative witness and the deferred quotient symbol, we write the former as y and reserve $Q_t[j]$ exclusively for deferred quotient witnesses.

For example:

- inverse row:
$$L = \widehat{y}\,\widehat{d}, \qquad D = \widehat{c}, \qquad \widehat{y}\,\widehat{d} = \widehat{c} + p\,\widehat{Q}_{\mathrm{inv}};$$
- $E \times F$ coordinate row:
$$L = \widehat{x}_k\,\widehat{a}, \qquad D = \widehat{c}, \qquad \widehat{x}_k\,\widehat{a} = \widehat{c} + p\,\widehat{Q}_{\mathrm{EF},k};$$
- $E \times E$ coordinate row:
$$L = \sum_{i,j=0}^{3} m_{ij}^k\,\widehat{x}_i\,\widehat{y}_j, \qquad D = \widehat{c}, \qquad \sum_{i,j=0}^{3} m_{ij}^k\,\widehat{x}_i\,\widehat{y}_j = \widehat{c} + p\,\widehat{Q}_{\mathrm{EE},k}.$$

Here m_{ij}^k denotes an audited non-negative normalized template coefficient in the instantiated library, not a raw signed basis coefficient.

To upgrade field equalities to integer equalities in a soundness proof, we require the range bound for *all satisfying witnesses* (not only honest ones).

Definition 1.11 (Row-range discipline)**.** *For each local relation t, there exists $B_t \in \{2^{31}, 2^{66}\}$ such that every satisfying row assignment obeys*
$$0 \le \widehat{L_t[j]} < pB_t + p < r.$$

Lemma 1.12 (Monomial no-wrap)**.** *Let $x, y \in R$ satisfy:*

(i) x is the image in R of an integer $\widehat{x}$ with $0 \le \widehat{x} \le U_x$;

(ii) y is the image in R of an integer $\widehat{y}$ with $0 \le \widehat{y} \le U_y$;

(iii) $U_x U_y < r$.

Then the field product $xy \in R$ is the image in R of the integer product $\widehat{x}\widehat{y}$, and $0 \le \widehat{x}\widehat{y} < r$.

Proof. In $\mathbb{Z}/r\mathbb{Z}$ we have $x \equiv \widehat{x}$ and $y \equiv \widehat{y}$, hence $xy \equiv \widehat{x}\widehat{y} \pmod{r}$. By (iii), $0 \le \widehat{x}\widehat{y} \le U_x U_y < r$, so reduction modulo r is trivial and xy is represented by $\widehat{x}\widehat{y}$ in R. □

Audited no-wrap bounds (moved to appendix). We discharge Definition 1.11 and the corresponding RHS no-wrap bounds for all satisfying witnesses in Appendix B.

Template convention (non-negative normalization). This manuscript restricts to row templates compiled into a *non-negative* deferred-quotient form $L_t = D_t + pQ_t$, where L_t and D_t are non-negative combinations of degree-≤ 2 monomials over ranged row variables and Q_t is bound to one of the declared non-negative quotient intervals (e.g., I_{31} or I_{66}). Handling row relations whose satisfying assignments force a negative Euclidean quotient (e.g., $L_t < D_t$ on some satisfying rows) would require signed quotient ranges or a different normalization scheme, and is outside the scope of this manuscript.

Worked normalization example. A simple illustration is a row relation initially written as

$$x_0 + x_1 - x_2 \equiv 0 \pmod{p},$$

where x_0, x_1, x_2 are designated residue cells. In the restricted model of this manuscript, such a relation is compiled to the non-negative deferred-quotient form

$$x_0 + x_1 = x_2 + pq,$$

with $x_2 \in I_p$ and $q \in \{0, 1\}$, because

$$0 \le x_0 + x_1 \le 2(p-1) < 2p.$$

This is the kind of normalization covered by the present theorem model. By contrast, families whose satisfying assignments force negative quotients or a signed remainder convention require a different normalization scheme and are outside the present scope.

Horner-parameter convention. Throughout this manuscript, the promoted-Horner parameter ρ is a fixed public extension-field constant $\rho \in \mathbb{F}_{p^4}$. It is not a witness cell and not an instance-dependent Fiat–Shamir challenge. Changing ρ requires regenerating the promoted-Horner template specialization and its audited bounds.

Theorem 1.13 (Core rowwise exactness from canonical lifts)**.** *Fix a satisfied row relation*

$$L_t[j] = D_t[j] + pQ_t[j] \quad \text{in } R.$$

Assume there exists $B_t \in \{2^{31}, 2^{66}\}$ and integers $\widehat{L_t[j]}, \widehat{D_t[j]}, \widehat{Q_t[j]}$ such that:

(i) $L_t[j]$, $D_t[j]$, and $Q_t[j]$ are the images in R of $\widehat{L_t[j]}$, $\widehat{D_t[j]}$, and $\widehat{Q_t[j]}$, respectively;

(ii)

$$0 \le \widehat{L_t[j]} < r;$$

(iii)

$$0 \le \widehat{Q_t[j]} < B_t;$$

(iv)

$$0 \le \widehat{D_t[j]} + p\,\widehat{Q_t[j]} < r.$$

Then

$$\widehat{L_t[j]} = \widehat{D_t[j]} + p\,\widehat{Q_t[j]} \quad \text{in } \mathbb{Z}.$$

If additionally $\widehat{D_t[j]} \in I_p$, then

$$\widehat{D_t[j]} = \widehat{L_t[j]} \bmod p, \qquad \widehat{Q_t[j]} = \left\lfloor \frac{\widehat{L_t[j]}}{p} \right\rfloor.$$

Proof. By assumption (i), the row values are the images in R of the integers $\widehat{L_t[j]}, \widehat{D_t[j]}, \widehat{Q_t[j]}$. Since

$$L_t[j] = D_t[j] + pQ_t[j] \quad \text{in } R,$$

we obtain

$$\widehat{L_t[j]} \equiv \widehat{D_t[j]} + p\,\widehat{Q_t[j]} \pmod{r}.$$

By assumptions (ii) and (iv), both integers

$$\widehat{L_t[j]} \qquad \text{and} \qquad \widehat{D_t[j]} + p\,\widehat{Q_t[j]}$$

lie in the interval $[0, r)$. Therefore they are equal in $\mathbb{Z}$:

$$\widehat{L_t[j]} = \widehat{D_t[j]} + p\,\widehat{Q_t[j]}.$$

If additionally $\widehat{D_t[j]} \in I_p$, uniqueness of Euclidean division by p yields

$$\widehat{D_t[j]} = \widehat{L_t[j]} \bmod p, \qquad \widehat{Q_t[j]} = \left\lfloor \frac{\widehat{L_t[j]}}{p} \right\rfloor.$$

□

Theorem 1.14 (Conditional rowwise no-vacuity under canonical binding)**.** *Assume the following four premises:*

(A) ***Canonical binding.*** *Residue, quotient, and carry columns are range-bound by the binding constraints* (1)–(3)*,* (4)–(5)*,* (6)–(7)*, and* (8) *or* (9)*. Canonical residue rows are exactly those rows whose $D_t[j]$ is a designated residue witness cell $c_t[j]$ with canonical lift $\widehat{c_t[j]} \in I_p$.*

(B) ***Audited no-wrap bounds.*** *For each active row relation (t, j) matched to family f, the audited bounds satisfy*

$$0 \le \widehat{L_t[j]} \le U_f^{\text{audit}} < pB_f + p < r$$

and

$$0 \le \widehat{D_t[j]} \le U_f^{D,\text{audit}}, \qquad U_f^{D,\text{audit}} + p(B_f - 1) < r.$$

.

(C) ***Declared template/realization coherence.*** *On every active row, the local realization expressions agree with the assigned audited normalized template as specified by Definition C.6 and discharged family-by-family in Appendix C.2 (Theorem C.8).*

(D) ***Complete wiring/coverage.*** *Every instantiated quotient/residue/carry slot is covered exactly once by the intended binding rules (no omissions), as formalized by Definitions 1.7, C.1, and C.3.*

Then for every satisfied row relation

$$L_t[j] = D_t[j] + pQ_t[j] \quad \textit{in } R, \tag{10}$$

the hypotheses of Theorem 1.13 hold. Consequently,

$$\widehat{L_t[j]} = \widehat{D_t[j]} + p\,\widehat{Q_t[j]} \quad \textit{in } \mathbb{Z}. \tag{11}$$

For canonical residue rows, $\widehat{c_t[j]}$ is the canonical remainder modulo p, and $\widehat{Q_t[j]}$ is the exact Euclidean quotient.

Proof. Fix any row relation (t, j) satisfying (10). By canonical binding together with Lemmas C.2 and C.4, every quotient/residue/carry cell participating in that row admits a canonical integer lift in its declared interval.

If $j \notin \Omega_t$, then Definition C.13 gives

$$L_t[j] = D_t[j] = Q_t[j] = 0,$$

so the hypotheses of Theorem 1.13 are immediate.

Assume now $j \in \Omega_t$. By Definition C.6 and Lemma C.10, the values $L_t[j]$ and $D_t[j]$ are the images in R of exact audited integer evaluations. By Lemma B.7,

$$0 \le \widehat{L_t[j]} < pB_t + p < r$$

for the class B_t of the matched family. By Lemma C.2 together with Definition B.2,

$$0 \le \widehat{Q_t[j]} < B_t.$$

By Lemma B.8,

$$0 \le \widehat{D_t[j]} + p\widehat{Q_t[j]} < r.$$

Thus all hypotheses of Theorem 1.13 hold, yielding (11). On canonical residue rows, the designated residue cell satisfies $\widehat{c_t[j]} = \widehat{D_t[j]} \in I_p$, so the final Euclidean-division statement follows from Theorem 1.13. □

Premise discharge map. For quick verification of Theorem 1.14, the four premises are discharged at:

Premise	Statement	Where discharged
(A)	Canonical binding constraints	Section 1.7, Lemmas C.2 and C.4
(B)	Audited no-wrap bounds	Appendix B, especially Lemma B.7 and Lemma B.8
(C)	Template/realization coherence	Appendix C.2, Theorem C.8
(D)	Complete wiring/coverage	Appendix C, Definitions 1.7, C.1, C.3

Mechanization status (separate from theorem statement). Theorem 1.14 is proved in the manuscript, including the familywise symbolic coherence closure in Appendix C.2. All mathematical statements used by the theorem are stated and proved in this manuscript. Separately, the released checker validates instantiated bound metadata, wiring/coverage metadata, row-activity metadata, and digest consistency for the current BN254 catalogue. No backend-independent cryptographic closure theorem is claimed here. For hor/car, this is explicitly a conditional theorem relative to the published template metadata.

1.10 Extension to a concrete Halo2-style backend (discussion)

The Tier-2 wrapper check itself can remain unchanged; typically it enforces the single compressed evaluation identity

$$\sum_t \lambda_t\big(l_t - d_t - pq_t\big) = h\, Z_\Omega(\zeta), \tag{12}$$

where

$$l_t = L_t(\zeta), \quad d_t = D_t(\zeta), \quad q_t = Q_t(\zeta), \quad h = H(\zeta),$$

and $\zeta \in R \setminus \Omega$ is a verifier challenge (sampled interactively or via Fiat–Shamir). The identification between compression-layer values $L_t(\omega^j), D_t(\omega^j)$ and rowwise values $L_t[j], D_t[j]$ is fixed by Definition 1.3.

Remark 1.15 (Backend-dependent closure boundary)**.** *Theorem 1.14 is the core algebraic exactness theorem in this manuscript. A fully closed cryptographic wrapper theorem additionally requires a separate backend proof of transcript binding/extraction, degree discipline for* (12)*, and Fiat–Shamir soundness. This manuscript does not prove such a backend-closure theorem. Full backend closure remains a separate proof obligation.*

1.11 Summary

The simple algebraic repair in this model to eliminate the p^{-1}-based vacuity is:

(1) retain the one-limb exact arithmetic over R,

(2) keep the single compressed polynomial evaluation check (12),

(3) add only a *native* canonical-binding layer consisting of:

- 31-bit lookup + one non-equality-to-p gate for each residue cell,
- 31-bit or 66-bit power-of-two lookup bounds for quotient cells,
- boolean or 3-bit lookup constraints for carries.

Within the restricted model and metadata premises above, this excludes the p^{-1}-based vacuity class and realizes the missing canonical-lift property guarantee $\mathcal{P}_{\text{canon}}$ at the row level while keeping Tier-2 arithmetic fully native over $\mathbb{F}_r$. The additional wrapper arithmetic-object formulas are deferred to Appendix A. Representative numeric instantiations are supplementary; the main qualitative comparison is Table 1 in Appendix A.

At claim level, the mechanized/non-mechanized split is exactly the one stated in Section 1.2; backend-level cryptographic closure remains outside the scope of this manuscript (Remark 1.15).

Claim boundary. The manuscript proves a restricted-model rowwise exactness theorem. For the hor/car families, the theorem is conditional on the published finite template and wiring metadata used by this instantiation (Appendices B and C). The released artifact provides reproducible instantiation checks (bounds/metadata/digest binding); a backend-specific transcript extraction and Fiat–Shamir theorem remains a separate proof layer.

A Supplementary Tier-2 cost accounting

Logical dependency note. This section reports symbolic arithmetic-object counts only (lookup cells, native multiplicative constraints, and linear recomposition constraints). It is not part of the rowwise exactness proof package, does not claim wall-clock performance, and does not assume lookup/multiplication cost-equivalence. The implemented Halo2 B1-lite comparison is supplementary and kept in Appendix D.

Let

$$\begin{aligned} N_p &:= \#\{\text{residue cells bound by (1)–(3)}\}, \\ N_{31} &:= \#\{\text{31-bit quotient cells bound by (4)–(5)}\}, \\ N_{66} &:= \#\{\text{66-bit quotient cells bound by (6)–(7)}\}, \\ N_{\text{bit}} &:= \#\{\text{boolean carry cells}\}, \\ N_8 &:= \#\{\text{3-bit carry cells}\}. \end{aligned}$$

We decompose quotient-cell totals by family as

$$N_{31} = N_{31}^{\text{EF}} + N_{31}^{\text{car}}, \qquad N_{66} = N_{66}^{\text{inv}} + N_{66}^{\text{EE}} + N_{66}^{\text{hor}},$$

where N_{31}^{car} is exactly the contribution of Q_{car} carry-normalization rows.

Then the added Tier-2 arithmetization cost is:

Lookup cells.

$$L_{\text{canon}} = 2N_p + 2N_{31} + 5N_{66} + N_8. \tag{13}$$

Native multiplicative constraints.

$$M_{\text{canon}} = N_p + N_{\text{bit}}. \tag{14}$$

These are:

- N_p non-equality constraints $(x - p)\nu_x = 1$,
- N_{bit} boolean constraints $c(c - 1) = 0$.

Linear recomposition constraints.

$$\text{Lin}_{\text{canon}} = N_p + N_{31} + N_{66}. \tag{15}$$

No limb-wise multiplication, no CRT consistency, and no foreign-field carry propagation appear anywhere in the canonical-binding layer.

A.0.1 Concrete promoted Horner row (scope-separated from carry-normalization rows)

Consider one promoted Horner row:

$$s = 2a_0 + a_1 + a_2 + a_3, \qquad s' = a_0 + \rho(a_1 + \rho(a_2 + \rho a_3)),$$

where ρ is the fixed Horner scalar parameter of the instantiated row family, with $\rho \in \mathbb{F}_{p^4}$ pinned for the instantiated promoted-Horner family (not a witness variable and not an instance-dependent Fiat–Shamir challenge). Multiplications by ρ are compiled into fixed specialized templates recorded in the public audit payload. With exact intermediate variables

$$m_3,\ h_2,\ m_2,\ h_1,\ m_1$$

and quotient/carry vectors as in the deferred-quotient arithmetization. This micro-count intentionally excludes separate carry-normalization rows. Although the underlying arithmetic corresponds to one Horner evaluation step of a degree-3 expression, the circuit realizes it via a fixed family of linear/quadratic subtemplates with intermediate witnesses; hence the audited template language remains degree-≤ 2. All Horner-related audit bounds and quotient-class assignments refer to these subtemplates, not to a single degree-3 monomial template. Then one promoted Horner row contains:

- 11 E-elements of canonical residues,

  $$a_0, a_1, a_2, a_3, s, s', m_3, h_2, m_2, h_1, m_1,$$

  hence

  $$N_p^{\text{round}} = 11 \cdot 4 = 44;$$

- 3 quotient vectors q_3, q_2, q_1, hence

  $$N_{66}^{\text{round}} = 3 \cdot 4 = 12;$$

- carry vectors $c_2, c_1, v \in \{0, 1\}^4$, hence

  $$N_{\text{bit}}^{\text{round}} = 12;$$

- one 3-bit carry vector $u \in \{0, \ldots, 4\}^4$, hence

  $$N_8^{\text{round}} = 4.$$

In this scope,

$$N_{31}^{\text{round}} = 0,$$

because no Q_{car} row is included.

Therefore the added binding counts per promoted Horner row are exactly:

$$L_{\text{canon}}^{\text{round}} = 2 \cdot 44 + 5 \cdot 12 + 4 = 152 \tag{16}$$

lookup cells,

$$M_{\text{canon}}^{\text{round}} = 44 + 12 = 56 \tag{17}$$

native multiplicative constraints, and

$$\text{Lin}_{\text{canon}}^{\text{round}} = 44 + 12 = 56 \tag{18}$$

linear recomposition constraints.

Carry-normalization add-on (31-bit quotient family). A carry-normalization row contributes one quotient vector $\{q_{\text{car}}^{(k)}\}_{k=0}^{3}$, i.e.

$$N_{31,\text{car}}^{\text{row}} = 4.$$

Hence each such row adds

$$\Delta L_{\text{canon}} = 2 \cdot 4 = 8, \qquad \Delta M_{\text{canon}} = 0, \qquad \Delta \text{Lin}_{\text{canon}} = 4.$$

If one carry-normalization row is paired with each promoted Horner row, the per-package counts are

$$L_{\text{canon}}^{\text{pkg}} = 160, \qquad M_{\text{canon}}^{\text{pkg}} = 56, \qquad \text{Lin}_{\text{canon}}^{\text{pkg}} = 60.$$

The carry-normalization add-on reuses the carry witnesses already counted in the paired promoted Horner row and introduces only the fresh quotient vector $\{q_{\text{car}}^{(k)}\}_{k=0}^{3}$ as additional bounded cells. Hence it contributes no extra boolean or 3-bit carry cost in this package accounting.

Let R_{hor} be the number of promoted Horner rows and R_{car} the number of carry-normalization rows. Then the total added binding overhead from these two row families is

$$L_{\text{canon}}^{\text{tot}} = 152\, R_{\text{hor}} + 8\, R_{\text{car}}, \tag{19}$$

$$M_{\text{canon}}^{\text{tot}} = 56\, R_{\text{hor}}, \tag{20}$$

and

$$\text{Lin}_{\text{canon}}^{\text{tot}} = 56\, R_{\text{hor}} + 4\, R_{\text{car}}. \tag{21}$$

These three equations are exact symbolic arithmetic-object counts for the added canonical-binding layer in this manuscript. This is consistent with (13)–(15) under

$$N_{66}^{\text{hor}} = 12 R_{\text{hor}}, \qquad N_{31}^{\text{car}} = 4 R_{\text{car}}.$$

Scope note for this section. Equations (19)–(21) are the main symbolic count outputs used in this note. Additional instantiated numeric tables are intentionally kept out of the main theorem narrative.

Binding primitive comparison (lookup limbs vs bit decomposition). As a backend-agnostic qualitative baseline at the cell level, we compare our lookup-limb binding against explicit bit decomposition. As a simple baseline, one can range-bind a k-bit value by explicit bit decomposition (introduce k boolean variables plus one recomposition constraint). Our binding mechanism replaces $\Theta(k)$ multiplicative constraints per cell with $O(1)$ lookup limbs and one recomposition, under the explicit premise that the Tier-2 backend provides a native lookup argument.

Cell type	Theorem instantiation (lookup-limb binding)	Bit-decomposition baseline
Residue cell in I_p	2 lookups + 1 linear + 1 non-equality	31 booleans + 1 linear + 1 non-equality
31-bit quotient cell in I_{31}	2 lookups + 1 linear	31 booleans + 1 linear
66-bit quotient cell in I_{66}	5 lookups + 1 linear	66 booleans + 1 linear
Boolean carry cell	1 boolean	1 boolean
3-bit carry cell in T_8	1 lookup	3 booleans + 1 linear

Table 1: Binding cost per cell (under native lookup availability): lookup limbs versus explicit bit decomposition.

Implementation note (supplementary B1-lite artifact only). For reproducibility, we report one implemented Halo2 circuit-shape comparison in Appendix D. This supplementary artifact is backend-specific, not task-equated, and not part of the main theorem claims or main quantitative evidence.

B Audited no-wrap bounds and template certificates

This appendix records the public audit certificate, deterministic bound verification, and instantiated constants used to justify the no-wrap premises in the rowwise exactness layer.

Row-family bound audit (all satisfying witnesses). To avoid treating Definition 1.11 as a standalone assumption, we publish the following row-family audit. Each row template is encoded in normalized non-negative deferred-quotient form and assigned a class $B_f \in \{2^{31}, 2^{66}\}$.

Row family f	Integer form for L_f	Certified bound U_f^{audit}	Class B_f
inverse	$L = y \cdot d$	$(2^{66}-1)(p-1)$	2^{66}
$E \times F$ coordinate	$L = \widehat{x}_k\,\widehat{a}$	$(p-1)^2$	2^{31}
$E \times E$ coordinate	$L = \sum_{i,j} m_{ij}^k \widehat{x}_i \widehat{y}_j$	$16\,(p-1)^3$	2^{66}
promoted Horner	family of linear/quadratic subtemplates (promoted Horner block)	$U_{\text{hor}}^{\text{audit}}$	2^{66}
carry normalization	linear carry-transfer form	$U_{\text{car}}^{\text{audit}}$	2^{31}

Here

$$M_{\text{EE}}^{\max} := \max_{k,i,j} \left|\widetilde{m_{ij}^k}\right| \le p-1 = 2147483646$$

is a conservative worst-case bound under canonical integer lifts (independent of basis coefficients).

Definition B.1 (Template-family audit certificate and deterministic bound compiler)**.** *A public row-audit certificate consists of a finite family index set $\mathcal{F}$. Each family $f \in \mathcal{F}$ has a finite template set $\mathcal{T}_f$. For each template $\tau \in \mathcal{T}_f$, the certificate provides:*

(i) a normalized non-negative left polynomial

$$L_{f,\tau}(\mathbf{z}) = \sum_{u \in \mathcal{L}_{f,\tau}} a_{f,\tau,u}\, z_u + \sum_{(u,v) \in \mathcal{B}_{f,\tau}} b_{f,\tau,uv}\, z_u z_v,$$

with coefficients $a_{f,\tau,u}, b_{f,\tau,uv} \in \mathbb{Z}_{\ge 0}$;

(ii) a normalized non-negative right polynomial

$$D_{f,\tau}(\mathbf{z}) = \sum_{u \in \mathcal{L}^D_{f,\tau}} \alpha_{f,\tau,u}\, z_u + \sum_{(u,v) \in \mathcal{B}^D_{f,\tau}} \beta_{f,\tau,uv}\, z_u z_v,$$

with coefficients $\alpha_{f,\tau,u}, \beta_{f,\tau,uv} \in \mathbb{Z}_{\ge 0}$;

(iii) interval bounds

$$0 \le z_u \le U_{f,u}^{\max}$$

induced by the class/range constraints;

(iv) the class $B_f \in \{2^{31}, 2^{66}\}$*;*

(v) deterministic per-template bounds

$$U_{f,\tau}^{L} := \sum_{u \in \mathcal{L}_{f,\tau}} a_{f,\tau,u}\, U_{f,u}^{\max} + \sum_{(u,v) \in \mathcal{B}_{f,\tau}} b_{f,\tau,uv}\, U_{f,u}^{\max} U_{f,v}^{\max},$$

$$U_{f,\tau}^{D} := \sum_{u \in \mathcal{L}_{f,\tau}^{D}} \alpha_{f,\tau,u}\, U_{f,u}^{\max} + \sum_{(u,v) \in \mathcal{B}_{f,\tau}^{D}} \beta_{f,\tau,uv}\, U_{f,u}^{\max} U_{f,v}^{\max}.$$

Define family-level maxima

$$U_f^{\text{audit}} := \max_{\tau \in \mathcal{T}_f} U_{f,\tau}^{L}, \qquad U_f^{D,\text{audit}} := \max_{\tau \in \mathcal{T}_f} U_{f,\tau}^{D}.$$

The compiler accepts only if each active instantiated row is syntactically matched to exactly one template (f, τ) *with* $\tau \in \mathcal{T}_f$*, while inactive row-table entries are handled separately by Definition C.13, and*

$$U_f^{\text{audit}} < pB_f + p, \qquad U_f^{D,\text{audit}} + p(B_f - 1) < r$$

holds for every f*. For the promoted-Horner and carry-normalization families in the audit table above, we write*

$$U_{\text{hor}}^{\text{audit}} := U_{f_{\text{hor}}}^{\text{audit}}, \qquad U_{\text{car}}^{\text{audit}} := U_{f_{\text{car}}}^{\text{audit}}.$$

$$U_{\text{hor}}^{D,\text{audit}} := U_{f_{\text{hor}}}^{D,\text{audit}}, \qquad U_{\text{car}}^{D,\text{audit}} := U_{f_{\text{car}}}^{D,\text{audit}}.$$

Definition B.2 (Family/quotient-class coherence)**.** *Whenever an active row is matched to template* (f, τ) *with family class* $B_f \in \{2^{31}, 2^{66}\}$*, every quotient slot appearing in that instantiated template has local class tag*

$$b_{s,\ell} = \begin{cases} 31, & B_f = 2^{31}, \\ 66, & B_f = 2^{66}. \end{cases}$$

Equivalently, every instantiated quotient value $Q_t[j]$ *in family* f *is wired to a column in* $\mathcal{Q}_{31}$ *if* $B_f = 2^{31}$*, and to a column in* $\mathcal{Q}_{66}$ *if* $B_f = 2^{66}$*.*

Symbolic certificate verifier relation (manuscript-level). Given the certificate data, the manuscript-level symbolic verifier relation requires the following checks:

(1) for each template (f, τ), parse $L_{f,\tau}$ and $D_{f,\tau}$, and ensure all certified coefficients are in $\mathbb{Z}_{\ge 0}$;

(2) compute $U_{f,\tau}^{L}$, $U_{f,\tau}^{D}$, then family maxima U_f^{audit}, $U_f^{D,\text{audit}}$ from Definition B.1;

(3) check

$$U_f^{\text{audit}} < pB_f + p, \qquad U_f^{D,\text{audit}} + p(B_f - 1) < r$$

for each family class B_f;

(4) verify the template map ω_T is total and functional on all active local scalar relation instances (s, ρ, j), that each such instance is matched to exactly one template (f, τ) with $\tau \in \mathcal{T}_f$, and that quotient-slot class tags are coherent with B_f as required by Definition B.2.

These checks define the manuscript-level symbolic certificate relation and are deterministic and independent of prover-provided runtime values.

Partially machine-checkable public certificate and checker. For this manuscript package we publish a concrete public certificate `certificates/public_certificate.json` and a deterministic checker `scripts/check_public_certificate.py`, yielding a partially machine-checkable certificate workflow. The public artifact repository is `https://github.com/junbyjun1238/zk2-bench-b1-lite`; the pinned artifact commit used for this manuscript is `https://github.com/junbyjun1238/zk2-bench-b1-lite/tree/2b1b28f7216bd85c75108caefc45dd1e437be061`. Within that pinned tree, the certificate, checker, and backend-instance files appear at `certificates/public_ce` `scripts/check_public_certificate.py`, and `certificates/h2dq_backend_instance.json`. The pinned public tree stores the manuscript under the filename `wrapper_note_option2.tex`, and the reproduction commands below follow that tree exactly. The checker deterministically:

(i) reconstructs per-template bounds and family maxima;

(ii) verifies the class/headroom checks $U_f^{\mathrm{audit}} < pB_f + p$, $U_f^{D,\mathrm{audit}} + p(B_f - 1) < r$;

(iii) verifies quotient/non-quotient coverage and selector-level wiring single-valued mapping/class consistency;

(iv) verifies relation-template reference integrity (`relation_id`$\mapsto$(`template_family`, `template_id`)), relation/row-activity metadata consistency, and inactive-row zero-extension flags;

(v) verifies digest binding between the certificate payload and the manuscript hash.

In this release, the deterministic checker does *not* mechanize full symbolic equality validation of declared local realization expressions against normalized template polynomials; that coherence relation remains specified as a published symbolic certificate relation (Definition C.6 and Lemma C.10). Reproduction command:

```
python scripts/check_public_certificate.py --certificate
certificates/public_certificate.json --manuscript
wrapper_note_option2.tex.
```

Pinned-instance check command:

```
python scripts/check_public_certificate.py --certificate
certificates/public_certificate.json --manuscript
wrapper_note_option2.tex --backend-instance
certificates/h2dq_backend_instance.json.
```

Backend-instance manifest role (reproducibility binding only). The optional backend-instance manifest binds the certificate payload digest, checker hash, manuscript hash, and fixed backend schedule metadata for reproducibility. In this manuscript, it is used only for artifact pinning and reproducible checker binding; it is not used as a standalone backend soundness theorem.

Instantiated bounds for the current template library (current manuscript).

family id	class bits	decimal left bound	decimal right bound
`inv`	66	158456324880954722595264004098	2147483646
`EF`	31	4611686009837453316	2147483646
`EE`	66	158456324585806817830375522176	2147483646
`hor`	66	158456324585806817830375522176	158456324585806817830375522176
`car`	31	4294967299	4294967299

For this manuscript draft, the table above is encoded exactly in `certificates/public\_certificate.json` (payload key `families`). Deterministic bound reconstruction is checked by `scripts/check\_public\_certificate.py`. In this instantiation, inverse/$E \times F$/$E \times E$ rows are canonical residue rows, so $\widehat{D_t[j]} = \widehat{c_t[j]} \in I_p$ and $U_{\mathrm{inv}}^{D,\mathrm{audit}} = U_{\mathrm{EF}}^{D,\mathrm{audit}} = U_{\mathrm{EE}}^{D,\mathrm{audit}} = p - 1$. In the current instantiated template library, the audited $E \times E$ family used for the constants above is treated in canonical non-negative form, so $\widehat{D_t[j]} = \widehat{c_t[j]} \in I_p$ on those rows. This note does not include alternative signed normalizations that would require signed quotient handling.

Lemma B.3 (Template-derived family bounds)**.** *Under Definition B.1, for every family $f \in \mathcal{F}$, every template $\tau \in \mathcal{T}_f$, and every satisfying assignment $\mathbf{z}$ of its row variables,*

$$0 \le L_{f,\tau}(\mathbf{z}) \le U_f^{\mathrm{audit}}.$$

and

$$0 \le D_{f,\tau}(\mathbf{z}) \le U_f^{D,\mathrm{audit}}.$$

Proof. All certified coefficients are non-negative and every variable is restricted to a non-negative interval. Therefore both $L_{f,\tau}$ and $D_{f,\tau}$ are coordinate-wise monotone on the certified box $\prod_u [0, U_{f,u}^{\max}]$. Their maxima are attained at the upper endpoint vector $(U_{f,u}^{\max})_u$, giving U_f^{audit} and $U_f^{D,\mathrm{audit}}$ by definition. Non-negativity gives the lower bounds. □

Concrete inequalities used in the audit. For $p = 2^{31} - 1$:

$$p \cdot 2^{66} + p = (2^{31} - 1)(2^{66} + 1) = 2^{97} + 2^{31} - 2^{66} - 1 < 2^{97}.$$

The audit verifies:

$$\begin{aligned}
&\text{inverse: } (2^{66} - 1)(p - 1) \\
&\qquad = 158456324880954722595264004098 \\
&\qquad < p \cdot 2^{66} + p, \\
&E \times F : (p - 1)^2 \\
&\qquad = 4611686009837453316 \\
&\qquad < p \cdot 2^{31} + p = 4611686018427387903, \\
&E \times E : 16\, M_{\mathrm{EE}}^{\max}(p - 1)^2 = 16(p - 1)^3 = 158456324585806817830375522176 < p \cdot 2^{66} + p.
\end{aligned}$$

The compiler also checks

$$U_{\mathrm{hor}}^{\mathrm{audit}} < p \cdot 2^{66} + p, \qquad U_{\mathrm{car}}^{\mathrm{audit}} < p \cdot 2^{31} + p.$$

and the RHS-side no-wrap conditions

$$\begin{aligned}
&U_{\mathrm{inv}}^{D,\mathrm{audit}} + p(2^{66} - 1) < r, \\
&U_{\mathrm{EF}}^{D,\mathrm{audit}} + p(2^{31} - 1) < r, \\
&U_{\mathrm{EE}}^{D,\mathrm{audit}} + p(2^{66} - 1) < r, \\
&U_{\mathrm{hor}}^{D,\mathrm{audit}} + p(2^{66} - 1) < r, \\
&U_{\mathrm{car}}^{D,\mathrm{audit}} + p(2^{31} - 1) < r.
\end{aligned}$$

Appendix-style instantiated audit values for selected templates. For the current wrapper template set, the promoted-Horner and carry-normalization family bounds are instantiated as follows.

Family	Instantiated template bound	Numeric value
promoted Horner	$\max\{16M_{\mathrm{EE}}^{\max}(p-1)^2,\ 2(p-1),\ 5(p-1)\}$	$16(p-1)^3 = 158456324585806817830375522176$
carry normalization	$\max\{2(p-1)+7,\ 2^{16}(2^{15}-1)+(2^{16}-1)\}$	$U_{\mathrm{car}}^{\mathrm{audit}} = 4294967299$

The promoted-Horner maximum comes from fixed-ρ multiplication sub-rows $m_3 = \rho a_3$, $m_2 = \rho h_2$, $m_1 = \rho h_1$. In this manuscript, $\rho \in \mathbb{F}_{p^4}$ is pinned at template-instantiation time, and these specialized subtemplates are conservatively audited under the same bound regime as the $E \times E$ coordinate family. The linear Horner sub-rows $h_2 = a_2 + m_3$, $h_1 = a_1 + m_2$, $s' = a_0 + m_1$, and $s = 2a_0 + a_1 + a_2 + a_3$ are strictly smaller.

The carry-normalization template family uses only low-width carry terms ($\{0,1\}$ and $\{0,\ldots,7\}$) and 31-bit limb-recomposition terms, hence the instantiated value above.

Definition B.4 (Instantiated full-family audit constants)**.** *For the current template set, define:*

$$U_{\mathrm{inv}}^{\mathrm{audit}} = (2^{66}-1)(p-1) = 158456324880954722595264004098, \tag{22}$$

$$U_{\mathrm{EF}}^{\mathrm{audit}} = (p-1)^2 = 4611686009837453316, \tag{23}$$

$$U_{\mathrm{EE}}^{\mathrm{audit}} = 16(p-1)^3 = 158456324585806817830375522176, \tag{24}$$

$$U_{\mathrm{hor}}^{\mathrm{audit}} = 16(p-1)^3 = 158456324585806817830375522176, \tag{25}$$

$$U_{\mathrm{car}}^{\mathrm{audit}} = 4294967299. \tag{26}$$

For RHS-side bounds, define:

$$U_{\mathrm{inv}}^{D,\mathrm{audit}} = p-1 = 2147483646, \tag{27}$$

$$U_{\mathrm{EF}}^{D,\mathrm{audit}} = p-1 = 2147483646, \tag{28}$$

$$U_{\mathrm{EE}}^{D,\mathrm{audit}} = p-1 = 2147483646, \tag{29}$$

$$U_{\mathrm{hor}}^{D,\mathrm{audit}} = 16(p-1)^3 = 158456324585806817830375522176, \tag{30}$$

$$U_{\mathrm{car}}^{D,\mathrm{audit}} = 4294967299. \tag{31}$$

The family-to-bound/class map is fixed as

$$(f_{\mathrm{inv}}, B_{f_{\mathrm{inv}}}, U_{f_{\mathrm{inv}}}^{\mathrm{audit}}) = (f_{\mathrm{inv}}, 2^{66}, U_{\mathrm{inv}}^{\mathrm{audit}}),$$

$$(f_{\mathrm{EF}}, B_{f_{\mathrm{EF}}}, U_{f_{\mathrm{EF}}}^{\mathrm{audit}}) = (f_{\mathrm{EF}}, 2^{31}, U_{\mathrm{EF}}^{\mathrm{audit}}),$$

$$(f_{\mathrm{EE}}, B_{f_{\mathrm{EE}}}, U_{f_{\mathrm{EE}}}^{\mathrm{audit}}) = (f_{\mathrm{EE}}, 2^{66}, U_{\mathrm{EE}}^{\mathrm{audit}}),$$

$$(f_{\mathrm{hor}}, B_{f_{\mathrm{hor}}}, U_{f_{\mathrm{hor}}}^{\mathrm{audit}}) = (f_{\mathrm{hor}}, 2^{66}, U_{\mathrm{hor}}^{\mathrm{audit}}),$$

$$(f_{\mathrm{car}}, B_{f_{\mathrm{car}}}, U_{f_{\mathrm{car}}}^{\mathrm{audit}}) = (f_{\mathrm{car}}, 2^{31}, U_{\mathrm{car}}^{\mathrm{audit}}).$$

$$(f_{\mathrm{inv}}, U_{f_{\mathrm{inv}}}^{D,\mathrm{audit}}) = (f_{\mathrm{inv}}, U_{\mathrm{inv}}^{D,\mathrm{audit}}),$$

$$(f_{\mathrm{EF}}, U_{f_{\mathrm{EF}}}^{D,\mathrm{audit}}) = (f_{\mathrm{EF}}, U_{\mathrm{EF}}^{D,\mathrm{audit}}),$$

$$(f_{\mathrm{EE}}, U_{f_{\mathrm{EE}}}^{D,\mathrm{audit}}) = (f_{\mathrm{EE}}, U_{\mathrm{EE}}^{D,\mathrm{audit}}),$$

$$(f_{\mathrm{hor}}, U_{f_{\mathrm{hor}}}^{D,\mathrm{audit}}) = (f_{\mathrm{hor}}, U_{\mathrm{hor}}^{D,\mathrm{audit}}),$$

$$(f_{\mathrm{car}}, U_{f_{\mathrm{car}}}^{D,\mathrm{audit}}) = (f_{\mathrm{car}}, U_{\mathrm{car}}^{D,\mathrm{audit}}).$$

Lemma B.5 (Instantiated full-family bound closure)**.** *Under Definition B.4,*

$$U_{\text{inv}}^{\text{audit}} < p \cdot 2^{66} + p, \tag{32}$$

$$U_{\text{EF}}^{\text{audit}} < p \cdot 2^{31} + p, \tag{33}$$

$$U_{\text{EE}}^{\text{audit}} < p \cdot 2^{66} + p, \tag{34}$$

$$U_{\text{hor}}^{\text{audit}} < p \cdot 2^{66} + p, \tag{35}$$

$$U_{\text{car}}^{\text{audit}} < p \cdot 2^{31} + p. \tag{36}$$

$$U_{\text{inv}}^{D,\text{audit}} + p(2^{66} - 1) < r, \tag{37}$$

$$U_{\text{EF}}^{D,\text{audit}} + p(2^{31} - 1) < r, \tag{38}$$

$$U_{\text{EE}}^{D,\text{audit}} + p(2^{66} - 1) < r, \tag{39}$$

$$U_{\text{hor}}^{D,\text{audit}} + p(2^{66} - 1) < r, \tag{40}$$

$$U_{\text{car}}^{D,\text{audit}} + p(2^{31} - 1) < r. \tag{41}$$

Proof. (32), (33), and (34) are exactly the inverse/$E \times F$/$E \times E$ inequalities already verified in the concrete audit block above.

For (35), by (25) and (24),

$$U_{\text{hor}}^{\text{audit}} = U_{\text{EE}}^{\text{audit}} < p \cdot 2^{66} + p.$$

For (36), from (26) and

$$p \cdot 2^{31} + p = 4611686018427387903,$$

we get

$$4294967299 < 4611686018427387903.$$

For RHS-side bounds, we use explicit instantiated values:

$$U_{\text{inv}}^{D,\text{audit}} + p(2^{66} - 1) = 158456324954741698892249694207 < 2^{98},$$

$$U_{\text{EF}}^{D,\text{audit}} + p(2^{31} - 1) = 4611686016279904255 < 2^{98},$$

$$U_{\text{EE}}^{D,\text{audit}} + p(2^{66} - 1) = 158456324954741698892249694207 < 2^{98},$$

$$U_{\text{hor}}^{D,\text{audit}} + p(2^{66} - 1) = 316912649540548516720477732737 < 2^{98},$$

$$U_{\text{car}}^{D,\text{audit}} + p(2^{31} - 1) = 4611686018427387908 < 2^{98}.$$

Since BN254 scalar r has 254 bits, $2^{98} < r$. Hence (37)–(41) follow. □

Corollary B.6 (Instantiated no-wrap intervals for all audited families)**.** *For every audited family $f \in \{f_{\text{inv}}, f_{\text{EF}}, f_{\text{EE}}, f_{\text{hor}}, f_{\text{car}}\}$, if a satisfying row obeys $\widehat{L}_f \le U_f^{\text{audit}}$, $\widehat{D}_f \le U_f^{D,\text{audit}}$, and $\widehat{Q}_f \in [0, B_f)$, then*

$$0 \le \widehat{L}_f < pB_f + p < r.$$

$$0 \le \widehat{D}_f + p\widehat{Q}_f \le U_f^{D,\text{audit}} + p(B_f - 1) < r.$$

Proof. Immediate from Definition B.4, Lemma B.5, and $\widehat{Q}_f \in [0, B_f)$. □

Lemma B.7 (Row-range discipline from audited templates)**.** *Assume:*

(i) Definitions C.5, C.6, and C.13 hold, and every active local scalar relation instance is matched to one audited template (f, τ) with $\tau \in \mathcal{T}_f$;

(ii) all input wires satisfy the class/range constraints from Definitions 1.7 and C.3, and (1)–(9)*;*

(iii) Definitions B.1 and B.4 hold, and Lemma B.5 applies.

Then Definition 1.11 holds for all satisfying witnesses. More precisely, for every row-table entry (t, j)*, there exists* $B_t \in \{2^{31}, 2^{66}\}$ *such that*

$$0 \le \widehat{L_t[j]} < pB_t + p < r.$$

If $j \in \Omega_t$*, one may take* $B_t = B_f$ *for the matched family* f*; if* $j \notin \Omega_t$*, the claim is immediate from Definition C.13.*

Proof. Fix any row-table entry (t, j).

If $j \notin \Omega_t$, then Definition C.13 gives $L_t[j] = 0$, so

$$0 \le \widehat{L_t[j]} = 0 < pB_t + p < r$$

for either choice $B_t \in \{2^{31}, 2^{66}\}$.

Now assume $j \in \Omega_t$. By (i), there is a unique active local scalar relation instance realizing (t, j), matched to some audited template (f, τ). By Lemma C.10,

$$\widehat{L_t[j]} = L_{f,\tau}(\widehat{\mathbf{z}}_j)$$

and

$$0 \le \widehat{L_t[j]} \le U_f^{\text{audit}}.$$

By (iii) and Corollary B.6,

$$U_f^{\text{audit}} < pB_f + p < r.$$

Therefore

$$0 \le \widehat{L_t[j]} < pB_f + p < r.$$

Setting $B_t := B_f$ gives the claim. □

Lemma B.8 (RHS no-wrap from audited templates)**.** *Assume:*

(i) all assumptions of Lemma B.7;

(ii) Definitions 1.7, C.1, and B.2 hold;

(iii) Lemma C.2 applies.

Then for each row relation $L_t[j] = D_t[j] + pQ_t[j]$*,*

$$0 \le \widehat{D_t[j]} + p\widehat{Q_t[j]} < r.$$

If $j \in \Omega_t$ *and the matched template family is* f*, then one may use that family's class* B_f*; if* $j \notin \Omega_t$*, the claim is immediate from Definition C.13.*

Proof. Fix (t, j).

If $j \notin \Omega_t$, then Definition C.13 gives $D_t[j] = Q_t[j] = 0$, so the claim is immediate.

Now assume $j \in \Omega_t$ and let f be the family of the matched audited template. By Lemma C.10,

$$0 \le \widehat{D_t[j]} = D_{f,\tau}(\widehat{\mathbf{z}}_j) \le U_f^{D,\text{audit}}.$$

By Lemma C.2, $Q_t[j]$ is range-constrained by its slot class. By Definition B.2, that slot class matches the family class B_f. Hence

$$\widehat{Q_t[j]} \in [0, B_f).$$

Therefore

$$0 \le \widehat{D_t[j]} + p\widehat{Q_t[j]} \le U_f^{D,\mathrm{audit}} + p(B_f - 1).$$

By Definition B.1 and Lemma B.5,

$$U_f^{D,\mathrm{audit}} + p(B_f - 1) < r.$$

Combining gives the claim. □

C Coverage and wiring certificates

This appendix records the explicit quotient inventory and selector-level wiring certificates as public synthesis metadata.

Serialization granularity note. In the released certificate schema, wiring maps are serialized at selector-slot granularity because the instantiated layout is row-invariant on active rows. Accordingly, row indices are omitted from wiring-map blocks and captured separately by `row_activity` metadata.

C.1 Coverage certificate (explicit inventory)

We materialize Definition 1.7 as a public inventory certificate. The certificate lists every quotient/carry family, its semantic origin, and its assigned range class.

Family	Semantic origin	Class	Reason / bound target
Q_{inv}	inverse rows $L = y \cdot d$	$\mathcal{Q}_{66}$	product can reach $\Theta(p \cdot 2^{66})$
$Q_{\mathrm{EF},k}$ $(k = 0, \ldots, 3)$	$E \times F$ coordinate rows	$\mathcal{Q}_{31}$	$L < p^2 < p \cdot 2^{31} + p$
$Q_{\mathrm{EE},k}$ $(k = 0, \ldots, 3)$	$E \times E$ coordinate rows	$\mathcal{Q}_{66}$	conservative high-range class
q_3, q_2, q_1	promoted Horner rows	$\mathcal{Q}_{66}$	promoted Horner sub-template normalization quotients
Q_{car}	carry-normalization rows	$\mathcal{Q}_{31}$	low-width carry transfer
c_2, c_1, v	boolean carry rows	$\mathcal{C}_{\mathrm{bit}}$	$c(c-1) = 0$
u	3-bit carry rows	$\mathcal{C}_8$	$u \in \{0, \ldots, 7\}$

Column-level class map (no omitted quotient symbol). The quotient inventory is explicitly decomposed as

$$\mathcal{Q} = \mathcal{Q}_{\mathrm{inv}} \sqcup \mathcal{Q}_{\mathrm{EF}} \sqcup \mathcal{Q}_{\mathrm{EE}} \sqcup \mathcal{Q}_{\mathrm{hor}} \sqcup \mathcal{Q}_{\mathrm{car}},$$

where

$$\begin{aligned}
\mathcal{Q}_{\mathrm{inv}} &:= \{q_{\mathrm{inv}}^{(k)} \mid k \in \{0,1,2,3\}\},\\
\mathcal{Q}_{\mathrm{EF}} &:= \{q_{\mathrm{EF}}^{(k)} \mid k \in \{0,1,2,3\}\},\\
\mathcal{Q}_{\mathrm{EE}} &:= \{q_{\mathrm{EE}}^{(k)} \mid k \in \{0,1,2,3\}\},\\
\mathcal{Q}_{\mathrm{hor}} &:= \{q_3^{(k)}, q_2^{(k)}, q_1^{(k)} \mid k \in \{0,1,2,3\}\},\\
\mathcal{Q}_{\mathrm{car}} &:= \{q_{\mathrm{car}}^{(k)} \mid k \in \{0,1,2,3\}\}.
\end{aligned}$$

Class assignments are fixed as

$$\mathcal{Q}_{31} = \mathcal{Q}_{\mathrm{EF}} \sqcup \mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{Q}_{66} = \mathcal{Q}_{\mathrm{inv}} \sqcup \mathcal{Q}_{\mathrm{EE}} \sqcup \mathcal{Q}_{\mathrm{hor}}.$$

Hence each promoted Horner row contributes exactly

$$|\mathcal{Q}_{\text{hor}}| = 12$$

cells in $\mathcal{Q}_{66}$, matching $N_{66}^{\text{round}} = 12$ for the Horner block. This scope excludes carry-normalization rows; their quotient cells are $q_{\text{car}}^{(k)} \in \mathcal{Q}_{\text{car}} \subset \mathcal{Q}_{31}$ and are accounted separately in N_{31}. Any quotient symbol not in this certificate is invalid and circuit synthesis rejects the instance.

Definition C.1 (Selector-level quotient wiring certificate)**.** *Let Σ be the set of enabled row selectors, and for each $s \in \Sigma$, let $\Lambda_Q(s)$ be the local quotient slots used by the row template of s. Let $\Omega_s \subseteq \Omega$ be the rows on which selector s is active. The public metadata includes a deterministic map*

$$\omega_Q : \{(s, \ell, j) \mid s \in \Sigma,\ \ell \in \Lambda_Q(s),\ j \in \Omega_s\} \to \mathcal{Q}.$$

The certificate is valid only if:

(i) ***Single-valued mapping:*** *each local quotient slot (s, ℓ, j) maps to exactly one global quotient column $\omega_Q(s, \ell, j)$;*

(ii) ***Class consistency:*** *each local slot has a fixed class tag $b_{s,\ell} \in \{31, 66\}$, and*

$$\kappa(\omega_Q(s, \ell, j)) = b_{s,\ell}$$

for all active rows $j \in \Omega_s$;

(iii) ***Exact coverage:***

$$\text{Im}(\omega_Q) = \mathcal{Q};$$

(iv) ***No undeclared symbol:*** *synthesis rejects any quotient symbol not in $\mathcal{Q}$.*

Lemma C.2 (No omitted quotient wire)**.** *Under Definitions 1.7 and C.1, every instantiated local quotient slot is range-constrained by exactly one of* (4)–(5) *or* (6)–(7)*.*

Proof. Fix an active local quotient slot (s, ℓ, j). By Definition C.1(i), it maps to a unique column $q := \omega_Q(s, \ell, j) \in \mathcal{Q}$. By Definition 1.7, q lies in exactly one of $\mathcal{Q}_{31}$ or $\mathcal{Q}_{66}$, and each class is tied to exactly one range-constraint system. Hence the slot is constrained by exactly one of the two systems above. By (iii)–(iv), no instantiated quotient can bypass this partition. □

Definition C.3 (Selector-level residue/carry wiring certificate)**.** *Let Σ be the set of enabled row selectors. For each $s \in \Sigma$, let $\Lambda_N(s)$ be the local non-quotient slots of the row template of s (residue or carry slots), and $\Omega_s \subseteq \Omega$ the rows where s is active. Each local slot (s, ℓ) has a fixed tag $\tau_{s,\ell} \in \{p, \text{bit}, 8\}$. The public metadata includes a deterministic map*

$$\omega_N : \{(s, \ell, j) \mid s \in \Sigma,\ \ell \in \Lambda_N(s),\ j \in \Omega_s\} \to \mathcal{C}_p \sqcup \mathcal{C}_{\text{bit}} \sqcup \mathcal{C}_8.$$

The certificate is valid only if:

(i) ***Single-valued mapping:*** *each local slot (s, ℓ, j) maps to exactly one global column $\omega_N(s, \ell, j)$;*

(ii) ***Tag consistency:***

$$\tau_{s,\ell} = p \Rightarrow \omega_N(s, \ell, j) \in \mathcal{C}_p, \quad \tau_{s,\ell} = \text{bit} \Rightarrow \omega_N(s, \ell, j) \in \mathcal{C}_{\text{bit}}, \quad \tau_{s,\ell} = 8 \Rightarrow \omega_N(s, \ell, j) \in \mathcal{C}_8;$$

(iii) ***Exact coverage by tag:***

$$\text{Im}(\omega_N|_{\tau=p}) = \mathcal{C}_p, \quad \text{Im}(\omega_N|_{\tau=\text{bit}}) = \mathcal{C}_{\text{bit}}, \quad \text{Im}(\omega_N|_{\tau=8}) = \mathcal{C}_8;$$

(iv) ***No undeclared symbol:*** *synthesis rejects any residue/carry symbol outside* $\mathcal{C}_p \sqcup \mathcal{C}_{\text{bit}} \sqcup \mathcal{C}_8$.

Lemma C.4 (No omitted residue/carry wire)**.** *Under Definition C.3, every instantiated local residue/carry slot is range-constrained by exactly one of* (1)–(3), (8), *or* (9).

Proof. Fix any active local non-quotient slot (s, ℓ, j). By Definition C.3(i), it maps to a unique column $W := \omega_N(s, \ell, j)$. By tag consistency (ii), W belongs to exactly one of $\mathcal{C}_p, \mathcal{C}_{\text{bit}}, \mathcal{C}_8$, and each class is tied to exactly one binding rule family: residue constraints (1)–(3), bit carry (8), or 3-bit carry lookup (9). By (iii)–(iv), no instantiated residue/carry slot can bypass this partition. □

Definition C.5 (Selector-level scalar-relation realization map)**.** *Let* Σ *be the set of enabled row selectors. For each* $s \in \Sigma$, *let* $\Lambda_R(s)$ *be the set of local scalar relation slots in selector template* s, *and let* $\Omega_s \subseteq \Omega$ *be the rows where* s *is active. Let* $\mathcal{I}_{\text{rel}}$ *be the global relation-index set for row tables* $(L_t[j], D_t[j], Q_t[j])_{(t,j)\in\mathcal{I}_{\text{rel}}\times\Omega}$.

The public metadata includes a deterministic map

$$\omega_R : \{(s, \rho, j) \mid s \in \Sigma,\ \rho \in \Lambda_R(s),\ j \in \Omega_s\} \to \mathcal{I}_{\text{rel}}.$$

The certificate is valid only if:

(i) ***Single-valued mapping:*** *each active local scalar relation instance* (s, ρ, j) *maps to exactly one relation index* $t = \omega_R(s, \rho, j)$;

(ii) ***Exact active coverage / no duplication:*** *for each* (t, j) *with* $j \in \Omega_t$, *there exists a unique active local scalar relation instance* (s, ρ, j) *such that* $\omega_R(s, \rho, j) = t$;

(iii) ***Realization rule:*** *for that unique* (s, ρ, j), *the row table values satisfy*

$$L_t[j] = \mathcal{L}_{s,\rho}(\mathbf{z}_j), \quad D_t[j] = \mathcal{D}_{s,\rho}(\mathbf{z}_j), \quad Q_t[j] = \mathcal{Q}_{s,\rho}(\mathbf{z}_j),$$

where $\mathbf{z}_j$ *is the row-local witness tuple and* $\mathcal{L}_{s,\rho}, \mathcal{D}_{s,\rho}, \mathcal{Q}_{s,\rho}$ *are the declared local realization expressions for slot* (s, ρ);

(iv) ***No undeclared relation:*** *synthesis rejects any active local scalar relation instance not covered by* ω_R.

Definition C.6 (Template/realization coherence)**.** *Let* Σ *be the enabled selector set from Definition C.5. The public metadata includes a deterministic map*

$$\omega_T : \{(s, \rho, j) \mid s \in \Sigma,\ \rho \in \Lambda_R(s),\ j \in \Omega_s\} \to \{(f, \tau) \mid f \in \mathcal{F},\ \tau \in \mathcal{T}_f\}.$$

For each active local scalar relation instance (s, ρ, j), *define*

$$t := \omega_R(s, \rho, j), \qquad (f, \tau) := \omega_T(s, \rho, j).$$

Let $\mathbf{z}_j$ *be the row-local field tuple used by the local realization expressions, and let* $\widehat{\mathbf{z}}_j$ *be the tuple of canonical integer lifts of its ranged entries. Coherence requires*

$$L_t[j] = \mathcal{L}_{s,\rho}(\mathbf{z}_j) = \big[L_{f,\tau}(\widehat{\mathbf{z}}_j)\big]_R,$$

$$D_t[j] = \mathcal{D}_{s,\rho}(\mathbf{z}_j) = \big[D_{f,\tau}(\widehat{\mathbf{z}}_j)\big]_R,$$

where $[\cdot]_R$ *denotes reduction modulo* r *followed by the canonical embedding into* R. *Quotient slots appearing in the same instantiated template must satisfy Definition B.2.*

Worked end-to-end example (an $E \times E$ coordinate row). Fix an active row $j^\star \in \Omega$ and a coordinate $k \in \{0,1,2,3\}$. Consider a local scalar relation instance $(s, \rho, j^\star)$ whose selector slot corresponds to the $E \times E$ coordinate relation for coordinate k, and let

$$t := \omega_R(s, \rho, j^\star), \qquad (f, \tau) := \omega_T(s, \rho, j^\star).$$

In this case we have $f = \mathrm{EE}$ and one can take $\tau = k$. Let the row-local witness tuple be

$$\mathbf{z}_{j^\star} = (x_0, \ldots, x_3,\ y_0, \ldots, y_3,\ c) \in R^9,$$

where canonical binding implies there exist integer lifts $\widehat{x}_a, \widehat{y}_b, \widehat{c} \in I_p$ for all $a, b \in \{0,1,2,3\}$. The declared local realization expressions for this slot take the concrete form

$$L_t[j^\star] = \mathcal{L}_{s,\rho}(\mathbf{z}_{j^\star}) = \sum_{a,b=0}^{3} m_{ab}^k\, x_a y_b, \qquad D_t[j^\star] = \mathcal{D}_{s,\rho}(\mathbf{z}_{j^\star}) = c,$$

with quotient value $Q_t[j^\star] = \mathcal{Q}_{s,\rho}(\mathbf{z}_{j^\star})$ wired to the family slot $Q_{\mathrm{EE},k}$. Template/realization coherence then means these match the audited normalized templates

$$L_{\mathrm{EE},k}(\widehat{\mathbf{z}}_{j^\star}) := \sum_{a,b=0}^{3} m_{ab}^k\, \widehat{x}_a \widehat{y}_b, \qquad D_{\mathrm{EE},k}(\widehat{\mathbf{z}}_{j^\star}) := \widehat{c},$$

via $L_t[j^\star] = [L_{\mathrm{EE},k}(\widehat{\mathbf{z}}_{j^\star})]_R$ and $D_t[j^\star] = [D_{\mathrm{EE},k}(\widehat{\mathbf{z}}_{j^\star})]_R$. The audit certificate bounds the coefficients by $0 \le m_{ab}^k \le M_{\mathrm{EE}}^{\max} \le p-1$ and assigns the family class $B_{\mathrm{EE}} = 2^{66}$, yielding the certified family bounds $U_{\mathrm{EE}}^{\mathrm{audit}} = 16(p-1)^3$ and $U_{\mathrm{EE}}^{D,\mathrm{audit}} = p-1$. Here m_{ab}^k denotes the audited non-negative template coefficient in the instantiated library, not a raw signed basis coefficient. Finally, the coverage/wiring certificate assigns $Q_{\mathrm{EE},k}$ to a $\mathcal{Q}_{66}$ column, so canonical binding gives $\widehat{Q_t[j^\star]} \in [0, 2^{66})$. This is the concrete "selector $\to$ realization $\to$ normalized (L, D, Q) $\to$ bound $\to$ quotient class" chain used by the exactness argument.

C.2 Finite familywise closure of the coherence obligation for the current library

We now discharge the remaining manuscript-level symbolic coherence obligation for the concrete instantiated family set

$$\mathcal{F}_{\mathrm{inst}} := \{\mathrm{inv}, \mathrm{EF}, \mathrm{EE}, \mathrm{hor}, \mathrm{car}\}.$$

This subsection does not add a new backend claim. Its purpose is narrower: it replaces the earlier informal statement that the remaining families are "structurally simpler" by a finite familywise catalogue and a direct family-by-family manuscript proof.

Definition C.7 (Current local realization catalogue)**.** *For the current manuscript instance, every active local scalar relation instance belongs to exactly one of the following families.*

(i) ***Inverse family.*** *The local ranged tuple is*

$$(y, d, c, Q_{\mathrm{inv}}),$$

with declared local realization expressions

$$\mathcal{L}_{\mathrm{inv}}^{\mathrm{loc}}(y, d, c) = yd, \qquad \mathcal{D}_{\mathrm{inv}}^{\mathrm{loc}}(y, d, c) = c, \qquad \mathcal{Q}_{\mathrm{inv}}^{\mathrm{loc}}(y, d, c, Q_{\mathrm{inv}}) = Q_{\mathrm{inv}}.$$

The corresponding audited normalized template is

$$L_{\mathrm{inv}}(y, d, c) = yd, \qquad D_{\mathrm{inv}}(y, d, c) = c.$$

(ii) $E \times F$ ***family.*** *For each coordinate $k \in \{0,1,2,3\}$, the local ranged tuple is*

$$(x_0, x_1, x_2, x_3, a, c, Q_{\mathrm{EF},k}),$$

with

$$\mathcal{L}^{\mathrm{loc}}_{\mathrm{EF},k} = x_k a, \qquad \mathcal{D}^{\mathrm{loc}}_{\mathrm{EF},k} = c, \qquad \mathcal{Q}^{\mathrm{loc}}_{\mathrm{EF},k} = Q_{\mathrm{EF},k},$$

and audited template

$$L_{\mathrm{EF},k} = x_k a, \qquad D_{\mathrm{EF},k} = c.$$

(iii) $E \times E$ ***family.*** *For each coordinate $k \in \{0,1,2,3\}$, the local ranged tuple is*

$$(x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3, c, Q_{\mathrm{EE},k}),$$

with

$$\mathcal{L}^{\mathrm{loc}}_{\mathrm{EE},k} = \sum_{i,j=0}^{3} m^k_{ij}\, x_i y_j, \qquad \mathcal{D}^{\mathrm{loc}}_{\mathrm{EE},k} = c, \qquad \mathcal{Q}^{\mathrm{loc}}_{\mathrm{EE},k} = Q_{\mathrm{EE},k},$$

and audited template

$$L_{\mathrm{EE},k} = \sum_{i,j=0}^{3} m^k_{ij}\, x_i y_j, \qquad D_{\mathrm{EE},k} = c.$$

(iv) ***Promoted-Horner family.*** *Every active promoted-Horner scalar relation is one of the finitely many subtemplates*

$$m_3 = \rho a_3, \qquad h_2 = a_2 + m_3, \qquad m_2 = \rho h_2, \qquad h_1 = a_1 + m_2, \qquad m_1 = \rho h_1, \qquad s' = a_0 + m_1, \qquad s = 2a$$

taken coordinatewise. For the three fixed-ρ multiplication subtemplates, the audited normalized coefficient lists are the pinned non-negative coefficient tensors serialized in the public certificate, and the local realization expressions use those same pinned coefficient tensors. For the four linear subtemplates, the audited normalized templates are exactly the displayed linear forms and the local realization expressions are those same linear forms.

(v) ***Carry-normalization family.*** *Let $\mathcal{T}_{\mathrm{car}}$ denote the finite list of active carry-normalization templates serialized in the public certificate. For each $\tau \in \mathcal{T}_{\mathrm{car}}$, the declared local realization expression is a normalized non-negative linear form*

$$\mathcal{L}^{\mathrm{loc}}_{\mathrm{car},\tau}(\mathbf{z}) = \sum_{u \in \mathcal{U}_\tau} \lambda_{\tau,u} z_u, \qquad \mathcal{D}^{\mathrm{loc}}_{\mathrm{car},\tau}(\mathbf{z}) = c_\tau, \qquad \mathcal{Q}^{\mathrm{loc}}_{\mathrm{car},\tau}(\mathbf{z}) = Q_{\mathrm{car},\tau},$$

where $\lambda_{\tau,u} \in \mathbb{Z}_{\geq 0}$, $\mathbf{z}$ ranges over the row's declared residue/carry tuple, and the coefficient vector $(\lambda_{\tau,u})_{u \in \mathcal{U}_\tau}$ is exactly the one recorded for the corresponding audited normalized template in the public certificate.

No active local scalar relation belongs to any family outside $\mathcal{F}_{\mathrm{inst}}$.

Theorem C.8 (Finite familywise closure of premise (C))**.** *Assume Definitions C.5, C.6, B.1, and C.7. Then premise* (C) *of Theorem 1.14 reduces to a finite familywise symbolic equality check. More precisely, Definition C.6 holds for every active local scalar relation instance in the current instantiated family set*

$$\mathcal{F}_{\mathrm{inst}} = \{\mathrm{inv}, \mathrm{EF}, \mathrm{EE}, \mathrm{hor}, \mathrm{car}\}.$$

Proof. Fix any active local scalar relation instance (s, ρ, j). By Definition C.7, it belongs to exactly one of the five families listed there.

If it is an inv, EF, or EE instance, then the declared local realization expression and the audited normalized template are literally the same displayed polynomial, with the same quotient slot attached. Hence Definition C.6 holds immediately.

If it is a hor instance, then it is one of the seven promoted-Horner subtemplates listed in Definition C.7(iv). For the three fixed-ρ multiplication subtemplates, the local realization uses exactly the same pinned non-negative coefficient tensor as the audited normalized template recorded in the public certificate. For the four linear subtemplates, the local realization is exactly the same displayed linear form as the audited normalized template. Thus coherence holds term-by-term for every promoted-Horner active relation.

If it is a car instance, then by Definition C.7(v) the local realization expression is exactly the same normalized non-negative linear form as the audited template for the corresponding $\tau \in \mathcal{T}_{\mathrm{car}}$, again with the same designated quotient slot. Hence coherence holds term-by-term.

These five cases exhaust the active family set. Therefore Definition C.6 holds for every active local scalar relation instance in the current library. □

Corollary C.9 (Status of the coherence step in the current release)**.** *For the current manuscript instance, premise* (C) *of Theorem 1.14 is closed at the manuscript level by Theorem C.8. The released checker does not mechanize this step; it verifies only the numerical and metadata parts described in Appendix B.*

Proof. The manuscript-level symbolic closure is exactly Theorem C.8. The checker scope is the one stated in Appendix B. □

Lemma C.10 (Exact evaluation of audited templates on active rows)**.** *Assume Definitions C.5, C.6, and B.1. Let (t, j) be an active row-table entry, let (s, ρ, j) be its unique realizing local scalar relation instance, and let (f, τ) be the unique audited template assigned to that instance. If all row-local inputs lie in their certified integer intervals, then*

$$L_t[j] = \big[L_{f,\tau}(\widehat{\mathbf{z}}_j)\big]_R, \qquad D_t[j] = \big[D_{f,\tau}(\widehat{\mathbf{z}}_j)\big]_R,$$

with

$$0 \le L_{f,\tau}(\widehat{\mathbf{z}}_j) \le U_f^{\mathrm{audit}}, \qquad 0 \le D_{f,\tau}(\widehat{\mathbf{z}}_j) \le U_f^{D,\mathrm{audit}}.$$

Consequently, whenever

$$U_f^{\mathrm{audit}} < r, \qquad U_f^{D,\mathrm{audit}} < r,$$

the field values $L_t[j]$ and $D_t[j]$ are the images in R of those exact integers, i.e.

$$\widehat{L_t[j]} = L_{f,\tau}(\widehat{\mathbf{z}}_j), \qquad \widehat{D_t[j]} = D_{f,\tau}(\widehat{\mathbf{z}}_j).$$

Proof. The displayed field identities are exactly the content of Definition C.6. By Lemma B.3,

$$0 \le L_{f,\tau}(\widehat{\mathbf{z}}_j) \le U_f^{\mathrm{audit}}, \qquad 0 \le D_{f,\tau}(\widehat{\mathbf{z}}_j) \le U_f^{D,\mathrm{audit}}.$$

Expand $L_{f,\tau}$ and $D_{f,\tau}$ as in Definition B.1: they are sums of linear terms $a\, z_u$ and quadratic terms $b\, z_u z_v$ with non-negative integer coefficients. Since $U_f^{\mathrm{audit}} < r$ and $U_f^{D,\mathrm{audit}} < r$, also the per-template bounds $U_{f,\tau}^L$ and $U_{f,\tau}^D$ are $< r$. By Definition B.1, every linear contribution $a_{f,\tau,u} U_{f,u}^{\max}$ and quadratic contribution $b_{f,\tau,uv} U_{f,u}^{\max} U_{f,v}^{\max}$ is a non-negative summand of $U_{f,\tau}^L$ or $U_{f,\tau}^D$, hence is $< r$. Lemma 1.12 therefore applies to each multiplication. For every nonzero quadratic term, $b_{f,\tau,uv} \ge 1$, so

$$U_{f,u}^{\max} U_{f,v}^{\max} \le b_{f,\tau,uv} U_{f,u}^{\max} U_{f,v}^{\max} < r.$$

Hence Lemma 1.12 applies first to $z_u z_v$, and then again to multiplication by $b_{f,\tau,uv}$. Zero-coefficient terms are omitted from the template representation. Thus every summand evaluates in R as the image of its exact integer value. Because all summands are non-negative and the total is $< r$, addition does not wrap either, so the field evaluations coincide with the images in R of the total integer evaluations. The canonical lifts are therefore exactly the displayed integers. □

Corollary C.11 (Exact audited evaluation is closed for the full current family set)**.** *In the current manuscript instance, Lemma C.10 applies to every active local scalar relation instance in*

$$\mathcal{F}_{\text{inst}} = \{\text{inv}, \text{EF}, \text{EE}, \text{hor}, \text{car}\}.$$

Proof. By Theorem C.8, the template/realization coherence premise required by Lemma C.10 holds for every active family in the current instantiated library. The remaining premises of that lemma are already family-agnostic. □

Lemma C.12 (No omitted active local relation)**.** *Under Definition C.5, every active local scalar relation instance contributes to exactly one row-table entry* $(L_t[j], D_t[j], Q_t[j])$*, and every active row-table entry arises from exactly one such instance.*

Proof. Immediate from Definition C.5(i)–(ii). □

Definition C.13 (Row-table activity partition and zero extension)**.** *For each relation index* t*, there is a public active-row set* $\Omega_t \subseteq \Omega$*, and row-table columns* $(L_t[j], D_t[j], Q_t[j])_{j\in\Omega}$*. The circuit enforces:*

(i) ***Active-row realization:*** *for each* $j \in \Omega_t$*, the tuple* $(L_t[j], D_t[j], Q_t[j])$ *is instantiated from exactly one active local scalar relation instance, consistent with Definition C.5;*

(ii) ***Inactive-row fixing:*** *for each* $j \in \Omega \setminus \Omega_t$*,*

$$L_t[j] = 0, \qquad D_t[j] = 0, \qquad Q_t[j] = 0.$$

Lemma C.14 (No inactive-row influence on compressed evaluation)**.** *Under Definitions C.13 and 1.3, for each relation index* t *and each* $j \in \Omega \setminus \Omega_t$*,*

$$L_t(\omega^j) - D_t(\omega^j) - pQ_t(\omega^j) = 0.$$

Consequently, any non-zero residual entry occurs only on active rows.

Proof. Fix t and $j \in \Omega \setminus \Omega_t$. By Definition C.13, $L_t[j] = D_t[j] = Q_t[j] = 0$. By Definition 1.3, $L_t(\omega^j) = L_t[j]$, $D_t(\omega^j) = D_t[j]$, and $Q_t(\omega^j) = Q_t[j]$. Substituting gives the claimed zero residual. □

D Supplementary implemented B1-lite circuit-shape comparison

This appendix records the implemented Halo2 harness comparison referenced in the main text. It remains a circuit-shape comparison under one backend, not a task-equated benchmark.

In this harness, B1-lite versus the repair fragment gives $352/72 = 4.89\times$ in rows and $256/56 = 4.57\times$ in multiplicative constraints; versus the Repair B2 paired package, it gives $352/76 = 4.63\times$ in rows and again $256/56 = 4.57\times$ in multiplicative constraints. The row counts are backend-specific physical-row counts and are not expected to be a simple function of symbolic arithmetic-object counts, because lookup batching and gate packing can place multiple

Metric	Repair fragment	Repair B2 paired package	B1-lite baseline
Rows	72	76	352
Lookup cells	152	160	126
Native multiplicative constraints	56	56	256
Linear constraints	56	60	63

Table 2: Implemented Halo2 circuit-shape comparison from the artifact harness. The B1-lite baseline includes explicit 16×16 limb cross products and carry-chain constraints for a 256-bit multiplication, but excludes modular reduction and CRT-consistency checks; thus it is a conservative partial baseline for the implemented schoolbook-limb wrong-field family.

objects in one row. The lookup-cell ratio goes in the opposite direction, matching the design goal of this repair layer: trading multiplicative constraints for batched native lookups.[1]

References

[1] Ariel Gabizon, Zachary J. Williamson, and Oana Ciobotaru. PLONK: Permutations over Lagrange-bases for Oecumenical Noninteractive arguments of Knowledge. IACR ePrint 2019/953, 2019. `https://eprint.iacr.org/2019/953`.

[2] Ariel Gabizon and Zachary J. Williamson. Plookup: A simplified polynomial protocol for lookup tables. IACR ePrint 2020/315, 2020. `https://eprint.iacr.org/2020/315`.

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[4] Miguel Ambrona, Denis Firsov, and Inigo Querejeta-Azurmendi. Efficient Foreign-Field Arithmetic in PLONK. IACR ePrint 2025/695, 2025. `https://eprint.iacr.org/2025/695`.

[5] Privacy Scaling Explorations. halo2wrong: wrong-circuit gadgets for Halo2 (including non-native arithmetic). GitHub repository, commit `3eaa2c4703a333e9695dc980bebf3445b7a9b141`. `https://github.com/privacy-scaling-explorations/halo2wrong/tree/3eaa2c4703a333e9695dc980bebf3445b7a9b141`.

[1] Artifact repository: `https://github.com/junbyjun1238/zk2-bench-b1-lite`. Pinned manuscript artifact commit: `https://github.com/junbyjun1238/zk2-bench-b1-lite/tree/2b1b28f7216bd85c75108caefc45dd1e437be061`. Relevant files are `src/main.rs`, `results.md`, `wrapper_note_option2.tex`, `certificates/public_certificate.json`, `scripts/check_public_certificate.py`, and `certificates/h2dq_backend_instance.json`. Reproduce by running `cargo run --release`. Exact artifact binding for this manuscript is given by the published certificate/manifests and checker command in Appendix B.